

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	03 October 2026
Team ID	SWTID-2026-6471
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Requirement & Architecture	USN-1	As a student, I can use EduGenie to get learning support.	2	High	Poojadevi
Sprint-1	Requirement & Architecture	USN-2	Define functional requirements for EduGenie.	1	High	Nathiya
Sprint-1	Requirement & Architecture	USN-3	Design the solution architecture.	3	High	Poojadevi
Sprint-1	Requirement & Architecture	USN-4	Select technologies and AI model.	3	High	Poojadevi
Sprint-1	Data Preparation	USN-5	Prepare sample educational content for testing	3	Medium	Luvina
Sprint-2	AI Features	USN-6	As a student, I can ask questions and get answers.	3	High	Nathiya
Sprint-2	AI Features	USN-7	As a student, I can get simple explanations of topics.	3	High	Nathiya
Sprint-2	AI Features	USN-8	As a student, I can generate quizzes for practice.	3	High	Luvina
Sprint-2	AI Features	USN-9	As a student, I can summarize study materials.	3	Medium	Luvina
Sprint-2	AI Features	USN-10	As a student, I can get a learning path for a topic.	5	Medium	Nathiya
Sprint-2	Backend Integration	USN-11	Integrate Google Gemini API with FastAPI.	5	High	Poojadevi

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-3	Frontend	USN-12	Create a simple web interface for EduGenie.	3	High	Luvina
Sprint-3	Frontend	USN-13	Add options for different learning features.	2	High	Luvina
Sprint-3	Frontend	USN-14	Display AI-generated responses to students.	3	High	Luvina
Sprint-3	Integration	USN-15	Connect frontend with backend modules.	5	High	Nathiya+Luvina
Sprint-4	Testing & Deployment	USN-16	Perform functional testing of all features.	3	High	Nathiya
Sprint-4	Testing & Deployment	USN-17	Fix errors and improve application performance.	3	Medium	Poojadevi
Sprint-4	Testing & Deployment	USN-18	Deploy and demonstrate the application locally.	2	High	All members

Sprint-wise Story Points

- **Sprint-1: $2 + 1 + 3 + 3 + 3 = 12$**
- **Sprint-2: $3 + 3 + 3 + 3 + 5 + 5 = 22$**
- **Sprint-3: $3 + 2 + 3 + 5 = 13$**
- **Sprint-4: $3 + 3 + 2 = 8$**

Total Story Points = $12 + 22 + 13 + 8 = 55$

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	12	6 Days	28-Sep-2026	02-Oct-2026	12	29 Oct 2022

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-2	22	6 Days	28-Sep-2026	02-Oct-2026	22	
Sprint-3	13	6 Days	28-Sep-2026	02-Oct-2026	13	
Sprint-4	8	6 Days	28-Sep-2026	02-Oct-2026	8	

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Total Story Points = 55

Number of Sprints = 4

Velocity = Total Story Points Completed ÷ Number of Sprints

Velocity = 55 ÷ 4 = 13.75

Team Velocity = 13.75 Story Points per Sprint

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

EduGenie Sprint Burndown Chart

Story points remaining across the four planned sprints.

